| Class | Adm No |
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Candidate Name: ______________________________

H

millennia institute

# 2015 Preliminary Examination 2
## Pre-University 3

**H2 Biology** **9648/03**

**Applications Paper and Planning Question** **21 September 2015**

**2 hours**

Additional Materials: Writing paper

**READ THESE INSTRUCTIONS FIRST**

**Do not open this booklet until you are told to do so.**

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.
You will lose marks if you do not show your working or if you do not use appropriate units.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use | |
| --- | --- |
| 1 | |
| 2 | |
| 3 | |
| 4 | |
| 5 | |
| Total | |

**This question paper consists of 20 printed pages**

*For Examiner's Use*

Answer **all** questions.

**1.** Spiders cannot be farmed, in contrast to silkworms, due to their aggressive behaviour and territorial nature. Collecting silk from webs is a time-consuming task. It took eight years to make a golden spider silk cape from 1.2 million golden orb webs. Therefore, biotechnological production of recombinant spider silks is the only practicable solution to harvest silks on a larger scale and to meet growing needs of medicine.

group of scientists tried to clone the *MaSp1*, which codes for the main protein (Spidroin) in the spider web silk, into *Escherichia coli.* Several restriction enzymes were considered for this experiment. The target sites for the restriction enzyme HaeIII, BamHI, HindIII and EcoRI are shown in Fig. 1.1. The arrows drawn in each sequence show where the enzyme cuts the DNA molecule.

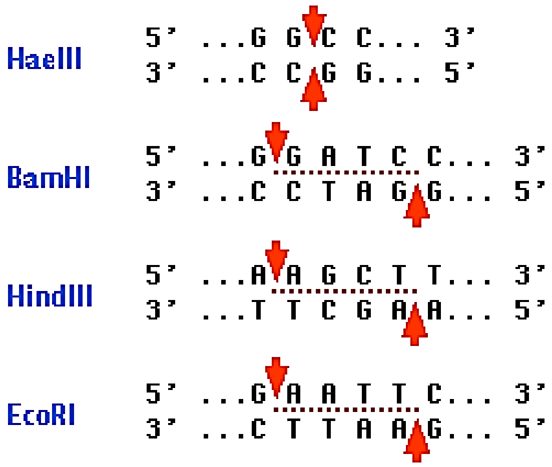


**Fig. 1.1**

**(a)** Explain why it is more suitable to use EcoRI than HaeIII for cloning.

*Eco*RI produces restriction fragments with sticky ends, allowing cut plasmid and cut gene of interest/pieces of DNA to anneal via complementary base pairing (and hydrogen bond formation);
*HaeIII* produces restriction fragments with blunt ends, hence requires addition of linker DNA;
which would need to be recognized, bind and cut by another enzyme to produce restriction fragments with sticky ends before annealing with the cut gene of interest, hence process takes longer/require more energy;
; for 1 mark, max is 2 marks

..........................................................................................[2]

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Their choice of plasmid is pGEM-7zf(+) vector which is commercially available. The multiple cloning sites (MCS) contain a number of restriction enzyme sites that can be utilised for cloning. Fig. 1.2 shows the restriction map of pGEM-7zf(+) plasmid.

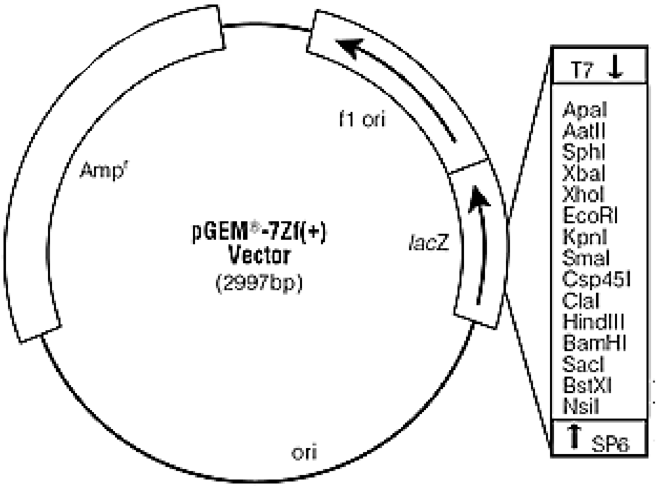


**Fig. 1.2**

Fig. 1.3 shows the base sequence in the MCS of the plasmid. Apart from the DNA sequence shown, there are no target sequences for BamHI, HindIII and EcoRI in the rest of the plasmid.

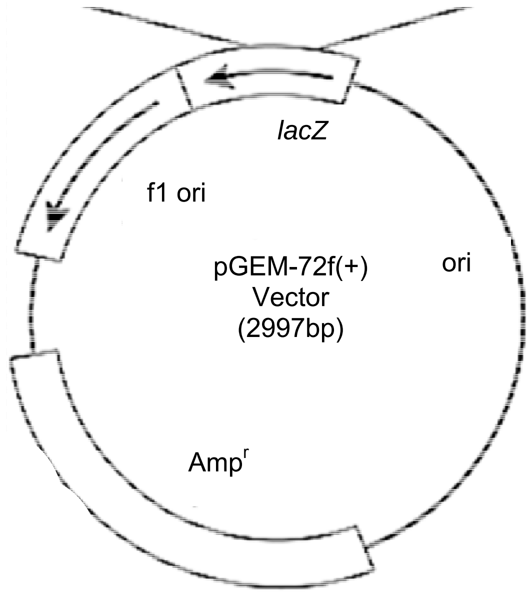


**Fig. 1.3**



**[Turn over**

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**(b)** With reference to Fig. 1.1 and Fig. 1.3, state the number of DNA fragments present after the plasmid has been cut with both BamHI and EcoRI together.

*2;*
; for 1 mark, max is 1 mark

..................................................[1]

To form recombinant DNA molecule, pGEM-7zf(+) and *MaSp1* were cleaved with both BamHI and EcoRI.

**(c)**

**(i)** Explain what is meant by recombinant DNA molecule.

It is a molecule that is made by combining DNA from two or more sources;
; for 1 mark, max is 1 mark

..................................................[1]

**(ii)** Suggest one reason why different restriction enzymes were used to create different sticky ends on each end of the *MaSpI* gene.

Allows the gene of interest to be inserted into the plasmid (cut by the same restriction enzymes) in the correct orientation;

; for 1 mark, max is 1 mark

**(iii)** Describe the subsequent steps involved in the formation of recombinant plasmid.

Use the same restriction enzymes, HindIII and BamHI, to digest DNA and plasmid to generate complementary sticky ends + The plasmid and the DNA are **mixed** together;
The sticky ends will **anneal** by complementary base pairing;
DNA ligase added to seal the nicks by catalyzing the formation of phosphodiester bonds between adjacent nucleotides;

; for 1 mark, max is 2 marks

**(d)** Explain how the scientists can use *lacZ* gene to select for bacteria containing the recombinant plasmid.

*lacZ* gene codes for functional β-galactosidase;
Bacteria are cultured in a medium containing X-gal;
a chromogenic substrate (colourless) that when hydrolysed / cleaved by β-galactosidase, will produce a blue product;

Colonies containing non-recombinant plasmids (bacteria containing the plasmid without inserted MaSpI gene at *lac Z*) will appear blue;
due to the intact *lacZ* gene (which codes for functional β-galactosidase)

Colonies containing recombinant plasmids (bacteria containing the plasmid with inserted gene at *lac Z*) will appear white as insertion of MaSpI gene disrupted the *lacZ* gene resulting in non-functional β-galactosidase produced

; for 1 mark, max is 3 marks

The scientists also explored the use of Polymerase Chain Reaction (PCR) to amplify the *MaSpI* gene. A pair forward and reverse primer was designed to amplify the *MaSpI* gene. Excess primers were added and the amplification of the target gene took place in vitro using a thermal cycler.

**(e)**

**(i)** Describe how PCR amplifies the *MaSpI* gene.

96°C + denatures / separates the sample DNA molecules Into 2 DNA strands;
Provide the template to synthesis a complementary daughter strand;
65°C + allows for the primers to anneal to the 3' ends of *MaSpI gene;*
72°C + allows for elongation of primer / addition of dNTPs;
Catalysed by the Taq polymerase;
To synthesis many copies of *MaSpI* gene;

; for 1 mark, max is 3 marks

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..............................................................................................

..............................................................................................[3]

**(ii)** Suggest why excess primers are added to the PCR mixture.

Prevents the re-annealing of the template DNA when the temperature decreases during the annealing and elongation stages;
More primers reduce the frequency of effective collision between the template strands;
; for 1 mark, max is 1 marks

**(iii)** Explain why a pair of primers is needed for PCR.

Double-stranded DNA is anti-parallel;
Taq polymerase can only synthesise DNA in 5' to 3' direction
Each primer binds to 1 strand of parental DNA;
Allowing for both strands to serve as template for replication;

; for 1 mark, max is 2 marks

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...[2]

[Total: 16]

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**2.** Cystic fibrosis (CF) used to be considered a fatal disease of childhood. With improved diagnosis and treatments to manage the disease, many people with CF now live well into adulthood.

CF is caused by mutation in the CFTR gene. The gene of interest can be found on the long arm of chromosome 7. Several restriction fragment length polymorphism (RFLP) markers located on the long arm of chromosome 7 have been found to be closely linked to CF.

One particular RFLP marker is shown in Fig. 2.1. The target sites of EcoRI are marked by arrows and the length of DNA between the targets sites are given in kilobases (kb). Most CF patients have a point mutation, which substitutes one of the bases at the restriction site marked with an asterisk (*).

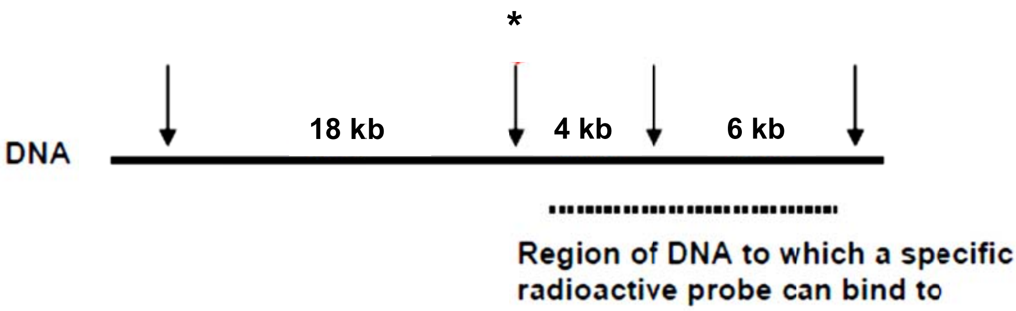


**Fig. 2.1**

CF can be diagnosed by cutting the DNA from patients with EcoRI and separate the DNA fragments by gel electrophoresis. The DNA fragments can then be transferred to nitrocellulose membrane and stained by a radioactive probe.

Fig. 2.2 is an autoradiograph which shows how CF can be detected by the altered size of the DNA fragments produced by cleavage with EcoRI. The banding patterns produced by DNA from three individuals, **A** to **C** are shown.

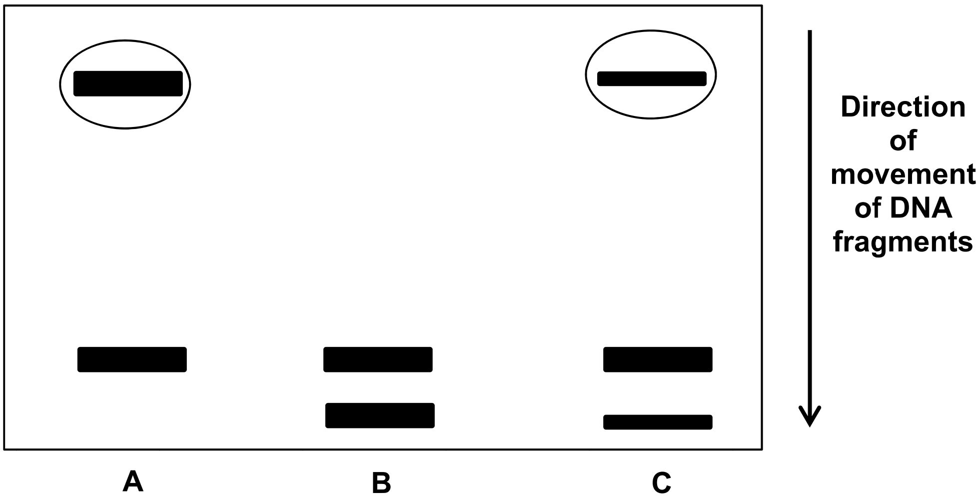


**Fig. 2.2**

For Examiner's Use

**(a)**

**(i)** Identify the DNA fragment of the mutant allele by circling the appropriate band(s) in Fig. 2.2. [1]

**(ii)** State the size of the DNA fragment identified in **(a)(i)**.

22kb;
; for 1 mark, max is 1 mark

[1]

**(iii)** With reference to Fig. 2.1 and 2.2, explain the banding pattern of individual **C**.

Individual C is heterozygous for CFTR gene locus + possess normal allele and mutant allele;
Digestion of normal CFTR allele gave rise to 2 fragments (4kb and 6kb)
Digestion of mutant allele gave rise to 2 fragment + 22kb and 6 Kb;
Thus, if you run it on a gel, it would a heterozygous would produce 3 bands + 4kb, 6kb & 22kb;
Thicker 6kb band in C because both mutant and normal allele will generate the 6 kb fragment upon EcoRI digestion;

; for 1 mark, max is 3 marks

[3]

**(b)** Suggest why a person without CF can also have the diseased RFLP marker.

During gamete formation, crossing over occurred at loci <u>between the RFLP marker and CF gene</u>;
Caused normal CF allele to be linked to diseased RFLP marker;

; for 1 mark, max is 1 marks

]

CF is an ideal candidate for gene therapy. Experimental trials have revealed that Lentivirus vectors of the retrovirus family show interesting properties for gene therapy. Lentivirus has a single stranded RNA genome with a reverse transcriptase enzyme.

**(c)** Describe two advantages of using Lentivirus vector in gene therapy.

Can target delivery of therapeutic allele to desired cells because LV can recognise specific receptors on target cells;
Therapeutic allele can be integrated into the chromosome, allowing for long term stability;
Therapeutic allele can be integrated into the chromosome, allowing for transfer to daughter cells during cell division;
Therapeutic allele can be integrated into the chromosome + do not require repeated treatments;

; for 1 mark, max is 2 marks

*For Examiner's Use*

**(d)** Describe the problems that may arise when a viral vector is used to introduce a functional allele into the target cell.

> Limited size of DNA insert due to viral capsid size;
> Insertional mutagenesis -> random insertion of DNA into genome and can disrupt other proteins;
> Viral vector regain virulence properties and evoke immune response;
>
> ; for 1 mark, max is 2 marks

Severe combined immunode”ficiency (SCID) is another ideal candidate for gene therapy. SCID is a potentially fatal primary immunodeficiency. There are at least 13 different genetic defects that can cause SCID. The most common form of SCID, affecting nearly 45% of all cases, is due to a mutation in a gene on the X chromosome. This form of SCID is also known as X-linked SCID.

**(e)** Explain why X-linked SCID is a suitable choice for gene therapy.

> SCID is a single gene disorder/ Mutation in 2 copies of IL2RG gene in females/ mutation in 1 copy of IL2RG gene in males gives rise to SCID;
> Only require one copy of dominant normal/functional allele is required to be inserted into target cell to restore normal phenotype;
>
> ; for 1 mark, max is 2 marks

[2]

**(f)** Describe the genetic basis of X-linked SCID.

> Mutation in the gene coding for IL-2 receptor gamma chain;
> Non-functional common gamma chain cause widespread defects in interleukin signalling;
> Interleukins and their receptors are involved in the development and differentiation of T and B cells;
> Complete failure of the immune system to develop and function, with low number or absent T cells and B cells;
>
> ; for 1 mark, max is 2 marks

A group of X-linked SCID patients underwent ex vivo retrovirus-mediated transfer of functional gene into $CD34^+$ bone marrow cells. Different dosages of viral vectors were administered for different patients. The bone marrow cells were then transferred back into the patients.

Efficacy of the gene transfer was determined by determining the presence of mRNA of the functional gene one week after the treatment. The results are shown in Table 2.1.

**Table 2.1**

| Patient | Gender | Age / years | Dose / pfu | Presence of mRNA of the functional gene | |
|---|---|---|---|---|---|
| | | | | Treated with empty viral vector | Treated with modified viral vector |
| 1 | F | 25 | $2 \times 10^7$ | No | No |
| 2 | M | 44 | $2 \times 10^7$ | No | No |
| 3 | M | 23 | $2 \times 10^8$ | No | No |
| 4 | M | 21 | $2 \times 10^8$ | No | No |
| 5 | F | 34 | $2 \times 10^9$ | No | Yes |
| 6 | M | 40 | $2 \times 10^9$ | No | No |
| 7 | F | 19 | $2 \times 10^9$ | No | Yes |
| 8 | F | 33 | $2 \times 10^{10}$ | No | Yes |
| 9 | M | 22 | $2 \times 10^{10}$ | No | No |

**(g)** Describe the relationship between the vector dosage and treatment efficacy.

No transcription of ADA gene in Patient 1 to 9 upon treatment with empty viral vector;
Transcription of functional gene only observed for high dose of viral vector above $2 \times 10^9$ pfu;
At high dose of viral vector, transcription of functional gene was only observed in females;
; for 1 mark, max is 2 marks.

........................................................................................................[2]

[Total: 16]

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**3.** Iron deficiency is the most common nutritional deficiency in the world. Use of genetic engineering to increase the iron content of the crops can be achieved by the introduction of genes coding for iron-binding proteins and iron-storage proteins.

FvC5sdp is an iron-binding protein expressed in fungus. Five transgenic plants (D-3, D-7, D-2, D-11 and D-12) expressing the gene coding for FvC5sdp were created. The total iron content of the fruits from wild type control and the transgenic lines were measured by atomic absorption spectroscopy. The results are shown in Fig. 3.1.

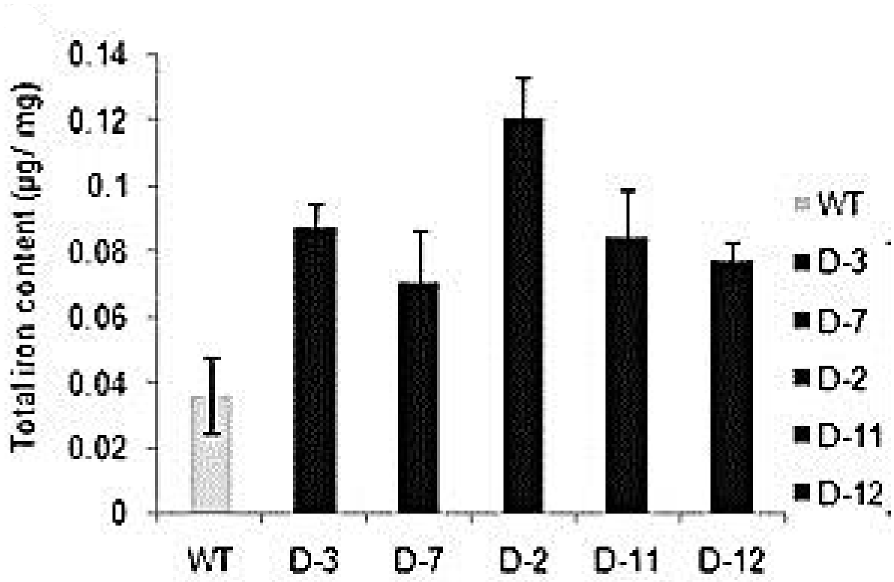


**Fig. 3.1**
(http://www.nature.com/srep/2012/121210/srep00951/full/srep00951.html)

**(a)** Explain the term “transgenic plants”.

Plants with a <u>foreign gene/gene from another species</u> inserted into all their cells;

; for 1 mark, max is 1 mark

**(b)** With reference to Fig. 3.1, account for the effect of introducing the transgene into the crops.

<u>Increase in total iron content of about 2–3 folds</u> were observed in all transgenic tomato fruits compared to that from control;
Since FvC5sdp is an iron-binding protein, it might result in an increased uptake of iron by the transgenic tomato plants;

; for 1 mark, max is 2 marks

[2]

Further investigations were carried out to determine if *FvC5sdp* gene taken from different species of fungus would enable maize plants to take up greater quantities of iron.

- *FvC5sdp* cDNA genes were obtained from four species of fungus (*Flammulina rossica*, *Flammulina similis*, *Flammulina stratosa* and *Flammulina velutipes)*. The cDNA genes were inserted into different plasmids.
- CaMV 35S promoter, a maize endosperm-specific promoter were also inserted into all of the plasmids.
- The four types of recombinant plasmids were then inserted into different cultures of maize cells.
- The iron content present in the maize plants was determined by atomic absorption spectroscopy.

The results are shown in Table 3.1.

**Table 3.1**

| **Source of *FvC5sdp* cDNA gene** | **Total Iron Content of Maize Plants / arbitrary units** |
|---|---|
| *Flammulina rossica* | 2 |
| *Flammulina similis* | 4 |
| *Flammulina stratosa* | 6 |
| *Flammulina velutipes* | 14 |

**(c)** Explain how the *FvC5sdp* cDNA was produced from each of the fungus species.

mRNA is isolated from cells that actively express the *FvC5sdp* gene;
Complementary DNA (cDNA) copy is synthesized using the enzyme reverse transcriptase;
Mixture treated with alkali / NaOH to digest mRNA;
DNA polymerase is used to make the cDNA of the gene of interest double-stranded / synthesize the 2nd DNA strand;

; for 1 mark, max is 2 marks

**(d)** Suggest why the CaMV 35S promoter was inserted into the plasmids.

To ensure that the *FvC5sdp* gene will be expressed in the endosperm of the maize plants;
A suitable promoter may not be present in the endosperm of the maize plants;

; for 1 mark, max is 1 mark

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**(e)** State **one** environmental risk of transgenic crops and **one** health risk from consuming these crops.

Risk to environment:
Gene transfer to wild-type relative or other related crops through cross-pollination. Upset the balance of ecosystem through hybridisation;
Gene transfer from crop to crop via bacteria or viruses with unknown effect. Upset the balance of ecosystem;
Recipient plant (plant that receives the gene) outcompetes natural variety or species / ref to ‘superweed’;
Transgenic crops expressing pesticide gene maybe toxic to non-pests;
Unknown long term effects on the environment / biodiversity;
; for 1 mark, max is 1 mark

Risk to humans:
Allergy;
Antibiotic resistance of gut bacteria if antibiotic marker used in the plasmid;
Long term toxicity;
Gene transfer from virus to human cells / gut bacteria, with unknown effect;
No long term studies done on effects to human health;

; for 1 mark, max is 1 mark

**4.** Pipistrellus is a genus of bats in the family Vespertilionidae. Classifying bat species based on their evolutionary relationship is difficult because of the large diversity of species. Fig. 4.1 shows the phylogenetic tree of the four known bat species (*Pipistrell pipistrellus*, *Pipistrell pygmaeus*, *Pipistrell javanicus* and *Pipistrellus abramus*).

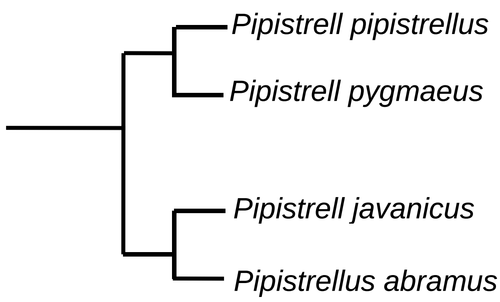



**Fig. 4.1**

Using this information and your own knowledge, design an experiment to show the molecular homology between four species of bats.

Your planning must be based on the assumption that you have been provided with the following equipment and materials.

- Tissue samples of the four bats species
- Pestle and mortar
- DNA extraction buffer solution
- Glass rods
- Microfuge tubes
- Centrifuge machine
- Restriction enzyme, HindIII
- Suitable reagents for polymerase chain reaction
- DNA primers
- Thermal cycler
- Agarose gel plate
- Suitable source of electrical current

Your plan should have a clear and helpful structure to include:

- An explanation of the theory to support your practical procedure,
- a description of the method used,
- relevant, clearly labelled diagram,
- relevant risk and precaution taken.
- The correct use of technical and scientific terms

[Total: 12]

**Theory ( 3 marks max)**
The evolutionary relationship between different bats species can be determined by comparing the similarity of DNA sequence of a common / conserved gene;
Closely related bats species will have very similar DNA sequence whereas distantly related species will have more different DNA sequences;
Gel electrophoresis may distinguish distantly related species if the difference in DNA sequences results in distinct sized DNA bands after PCR;
This could be due to deletion or duplication of specific section of the gene;
Differences in base sequences may result in the formation or deletion of a restriction site which will result in different restriction fragments after digestion with the appropriate restriction enzyme HindIII;

**Procedure (7 marks max)**
Extraction of DNA
1) **10g** of bats tissue samples from the 3 species were **homogenized** separately using **pestle and mortar**.
2) Add **10ml DNA extraction buffer solution** to the homogenized tissue. Stir the mixture using glass rod;

Preparation of DNA
3) Add **1ml of the mixture** is added to a centrifuge tube and the mixture is **centrifuged**.
4) The **supernatant containing DNA** is then transferred to a fresh centrifuge tube;

Amplification of DNA
5) Use a thermal cycler to amplify the common gene of the 4 species using **appropriate set of DNA primers flanking the common DNA sequence;**

| PCR step / *cycle* | Temperature / °C | Duration / s [1] |
|---|---|---|
| Denaturation | 95 | 60 |
| Annealing | 50 | 60 |
| Extension | 72 | 60 |
| Total *30* cycles<br>Student will be awarded mark for stating the number of cycle and duration for each stage of PCR; | | |

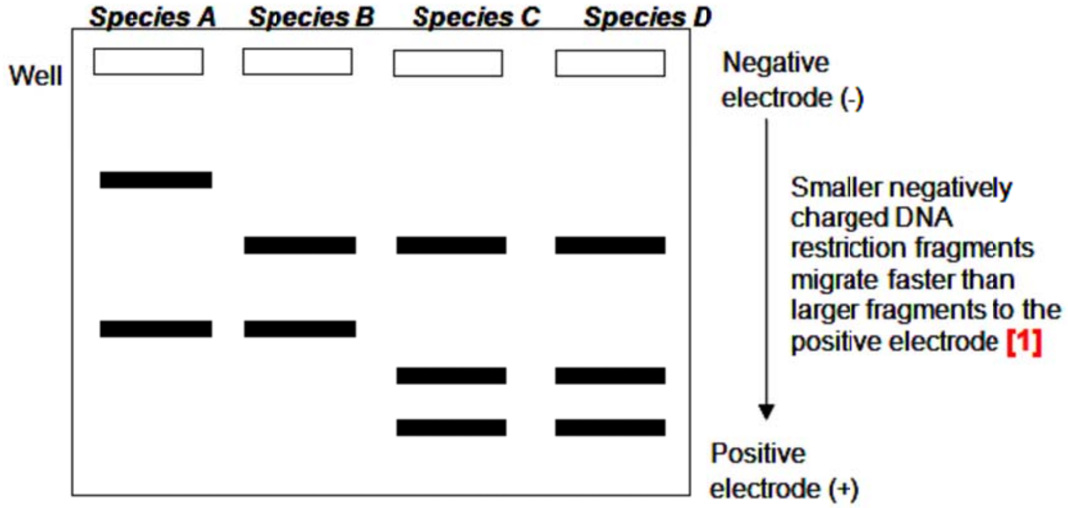
| Reagents per PCR tube | Volume / µl |
|---|---|
| DNA Taq polymerase mix including free DNA nucleotides | 20 |
| Forward DNA primer (1µM) | 5 |
| Backward DNA primer (1µM) | 5 |
| Buffer | 5 |
| DNA template (supernatant) | 5 |

Student will be awarded mark for stating all the reagents needed and stating the volume of each reagent;

Digestion of extracted DNA
6) Digest the **10µl** amplified DNA with **1µl HindIII** in a **37°C** incubator for **1h**;

Gel electrophoresis
7) Prepare 1% agarose gel. Pour the molten gel into gel casting tray. Insert the comb and let the gel set for 30 minutes.
8) Add **1µl loading dye** to **10µl of digested DNA**. Dispense the DNA samples into the wells of the gel;

9) Run at 100V for 1h.
10) Soak the gel in a solution of **ethidium bromide** and view the gel under **UV light** to compare the band patterns of the different species;



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Hypothetical results – drawn to depict correct evolutionary relationship or clearly described;

**Ethidium bromide (EtBr) is a carcinogen** which is **harmful** to the body. **Avoid direct skin contact** with the chemical by **wearing gloves** while handling the EtBr-stained gel;

; for 1 mark, max is 12 marks

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For Examiner's Use

**Free-response question**

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

**5.**

**(a)** Describe the use of restriction fragment length polymorphism analysis in creating a linkage map. [6]

*Using RFLP with known physical location on a chromosome;*
*Construct map using recombination frequency to determine order/sequence of genes and relative distance between different genes and genetic markers along a chromosome;*

*; for 1 mark, max is 1 mark*

*A linkage map shows the relative location;*
*or the order of genes along a chromosome;*
*constructed based on the assumption that the probability of a crossover between two genetic loci/RFLP markers is proportional to the distance separating the loci;*
*Linkage maps are usually constructed with several thousand known genetic markers spaced evenly throughout the genome;*
*The RFLP markers can be any genes or any other identifiable DNA sequences, such as variation number tandem repeats (VNTR) and short tandem repeats (STR);*

*; for 1 mark, max is 2 marks*

*How to carry out linkage mapping using RFLP:*
*Linkage map must be done based on experimental crosses;*
*After mating/fertilization, RFLP from parental and offspring organisms are obtained and analysed;*
*Using gel E & Southern blot/nucleic acid hybridisation;*
*RFLP pattern obtained are used to calculate the total percentage of recombinant offspring;*
*Give an indication of the distance between the RFLP markers/sequences based on the recombination frequencies obtained;*
*the farther apart the two RFLP markers/sequences are, the higher the probability that a crossover that generates genetic variation will occur between them and therefore the higher the recombination frequency;*
*For example, if 70% of the progeny produced are parental and 30% were recombinant, the two RFLP loci are 30 centi-Morgans (cM) apart from each other;*

*; for 1 mark, max is 3 marks.*

**(b)** Describe the unique features of stem cells and explain normal functions of embryonic stem cells in a living organism. [6]

*Unique features of stem cells*
*Unspecialized cells and lack tissue specific structures;*
*Able to proliferate to self-renew indefinitely by mitotic cell division*
*Able to differentiate under appropriate conditions dependent on the specific extracellular and intracellular signals;*

*Normal Function of stem cells*
*ES cells have a function in growth of the entire organism from the blastocyst stage;*
*ES can divide continuously to create all organs and specialised tissues in the entire organism;*
*ES cells are pluripotent;*
*Ability to differentiate to give rise to the 3 germ layers – endoderm, mesoderm and ectoderm;*

*; for 1 mark, max is 6 marks*

For Examiner's Use

**(c)** Using named plant and animal examples, discuss the significance of genetic engineering in improving the quantity of food supply [8]

*Plant*

*Bt Corn: genetically modified corn with the insertion of a gene from Bacillus thuringiensis into the corn;*
*This gene codes for Bt delta endotoxin;*
*The endotoxin kills a common pest that destroys corn crops;*
*Farmers growing Bt corn do not need to spray insecticides on their crops to get rid of the pests;*
*Since the crops are not infested with pests, this allows the farmers to increase their yields, productivity, improve their livelihoods and alleviate poverty;*

*Animal*

*Genetically Modified Salmon/ The Atlantic salmon has been genetically modified with two DNA sequences;*
*They are:*
*A promoter of an anti-freeze gene from an ocean pout.*
*A growth hormone gene from Chinook salmon;*
*These 2 DNA sequences enable the salmon to continue producing growth hormone all year round, even during the winter months;*
*GM salmon can grow 5 times faster than its wild type;*
*shorter maturation time, GM salmon raised in tanks can be harvested and sold in a shorter time and thus increasing the profit;*

; for 1 mark, max is 8 marks

[Total: 20]

**End of Paper**